

PHAROS

A Physics-Hybrid, Attention-Routed Mixture-of-Experts Architecture for RNA Structure and Contact-Map Prediction

Architecture Design Report — Version 0.1

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Abstract

RNA three-dimensional structure prediction has not had its AlphaFold2 moment. The best current methods reach a mean TM-score of approximately 0.55 on non-redundant benchmarks, which is the threshold for recovering roughly the correct fold, against routine scores above 0.9 for proteins. This report argues that the bottleneck is not model scale but *inductive bias*, and specifies an architecture, **PHAROS**, built around that claim.

Three empirical findings drive the design. First, ERNIE-RNA (86M parameters) outperforms RiNALMo (650M) and UNI-RNA (400M) on secondary-structure prediction, and nearly doubles cross-family generalisation relative to RiNALMo ($F_1 = 0.646$ versus 0.355), on the strength of a single hand-set, three-valued base-pairing prior added to its attention logits. Second, we measure on 117 non-redundant RNA chains that contacts per nucleotide *saturates* at 4.4–4.9 while contact-map density falls from 6.5% at $L \approx 56$ to 0.353% at $L \approx 2900$: RNA contact maps scale as $O(L)$, not $O(L^2)$, so a dense pair representation spends over 99% of its compute on empty space and would require 309 GB of activations at $L = 2048$. Third, we measure a monotonic 1.76σ gradient in normalised crystallographic B-factor as a function of distance to the nearest Mg^{2+} ion across 24,623 nucleotides, establishing that ion coordination and local rigidity are not independent features but two readouts of one phenomenon.

PHAROS accordingly (i) accepts **ionic conditions as a model input**, which no existing RNA structure predictor does, propagating them to attention logits through a Manning-renormalised screened-Coulomb bias; (ii) replaces the dense pair track with an **adaptive sparse pair track** retaining $K = 32L$ candidate edges, giving sixfold headroom over the measured contact requirement at 1.56% of dense cost; (iii) routes a **sparse mixture-of-experts** on local structural regime rather than sequence context, supported by our negative result that GNRA k -mer context predicts rigidity at only 0.073σ ; and (iv) supplies motif geometry through a **frozen retrieval bank** of 667 RNA 3D Motif Atlas classes rather than expecting the model to learn it from 6,661 structurally-resolved sequences. We also identify two supervision channels— Mg^{2+} sites and B-factor rigidity—that are already present in every mmCIF file the field downloads and are currently used by no structure predictor.

This is a design specification. No model has been trained. All quantitative claims are either marked **[measured]** and reproducible from the scripts in this repository, or cited to the literature.

Contents

1	Problem statement	3
1.1	The accuracy gap	3
1.2	Why RNA lags	3
2	The central argument: bias beats scale	3

3	Empirical analysis	4
3.1	RNA contact maps scale linearly, not quadratically	4
3.2	Ion coordination and local rigidity are one phenomenon	5
3.3	Ion inventory and coordination chemistry	5
3.4	A negative result that constrains the design	6
4	The PHAROS architecture	6
4.1	Inputs	6
4.2	Token trunk: hybrid attention with MoE	6
4.2.1	Physics-biased attention	7
4.2.2	Mixture-of-experts feed-forward	9
4.2.3	Parameter budget	9
4.3	Adaptive sparse pair track	9
4.4	Motif retrieval bank	11
4.5	Physics module	11
4.6	Heads and decoder	12
5	Data mapping and the acquisition gap	12
6	Novelty claims	13
7	Risks and open questions	13
8	Status	15
A	Reproducing the measurements	15

1 Problem statement

1.1 The accuracy gap

Table 1 gives the state of the art on a non-redundant RNA test set as of November 2025.

Table 1: Current state of the art in RNA 3D structure prediction.

Method	Mean TM-score	Approach
trRosettaRNA	0.548	MSA $\rightarrow$ restraints $\rightarrow$ folding
NuFold	0.525	AlphaFold2-style nucleotide frames
AlphaFold3 (server)	0.510	all-atom diffusion
AlphaFold3 (local)	0.504	all-atom diffusion

A TM-score near 0.5 means the topology is approximately right and little else. The CASP16 signal is equally instructive: leading RNA groups won by integrating AlphaFold3 *into physics-based modelling* rather than by scaling a neural network. Physics-plus-machine-learning hybrids are currently ahead in RNA, which is the premise this architecture is built on.

1.2 Why RNA lags

1. **Data scarcity.** Our catalogue records 6,661 unique sequences with experimental 3D structure against 1.37 billion sequences, a coverage of 0.0005%. Proteins had roughly 220,000 structures at the AlphaFold2 moment.
2. **Redundancy within that data.** 15,441 RNA3DB chains collapse to 3,157 unique sequences; 5S rRNA alone accounts for 5,976 BGSU equivalence classes.
3. **Family coverage.** Only 100 of 4,227 Rfam families (2.4%) have any 3D representative.
4. **Conformational plasticity.** Riboswitches are *defined* by adopting multiple functional folds; single-structure prediction is the wrong target for a large fraction of functional RNA.
5. **Ion dependence.** RNA is a polyanion. Its fold is not defined without specifying the ionic environment, yet no mainstream predictor accepts ionic conditions as an input.
6. **Target noise.** Median resolution in our catalogue is 3.10 Å, with only 13% of chains at 2.5 Å or better.

2 The central argument: bias beats scale

Table 2 contains the most important result in the recent RNA language-model literature.

Table 2: ERNIE-RNA versus larger dense models. A $7.5\times$ parameter deficit is overturned by one hand-set prior.

Model	Params	bpRNA-1m F_1	bpRNA-new F_1
ERNIE-RNA (attention map)	86M	0.873	—
ERNIE-RNA (embedding)	86M	0.862	0.646
RiNALMo	650M	0.850	0.355
UNI-RNA	400M	0.821	—
RNA-FM	100M	0.808	—

ERNIE-RNA’s entire architectural advantage is an additive bias on attention logits. In its first layer,

$$\text{Attention}(Q, K) = \frac{QK^\top}{\sqrt{d}} + B_{\text{pair}}(i, j), \quad \text{Attention}_l(Q, K) = \frac{QK^\top}{\sqrt{d}} + \text{Attention}_{l-1}(Q, K), \quad (1)$$

with B_{pair} set by hand to 3 for C–G, 2 for A–U, a tunable $\alpha \approx 0.8$ for G–U, and 0 on the diagonal.

The cross-family result is the decisive one: 0.646 versus 0.355. A fixed physical prior nearly doubles generalisation to unseen families, because the prior is true regardless of family whereas learned family statistics are not.

Thesis. If a crude three-valued prior buys more than $7\times$ the parameters, the marginal return on *better physics* exceeds the marginal return on *more scale*. PHAROS replaces that hand-set prior with a learned, ion-aware, motif-aware, coevolution-aware one.

Its limitations define our openings. The prior is static, so a G–U pair inside a kink-turn is treated identically to one in a free helix. It is nearest-neighbour-blind, whereas real pairing energy is stacking-dependent. It encodes only canonical pairing, while non-canonical pairs define RNA 3D motifs and carry most tertiary information. And it ignores ions entirely.

3 Empirical analysis

All results in this section were measured for this report on 180 BGSU non-redundant representative structures at ≤ 3.0 Å, downloaded and analysed by scripts in `scripts/sampling/`. Raw outputs are in `data/samples/analysis/`.

3.1 RNA contact maps scale linearly, not quadratically

Contacts are defined as nucleotide pairs with any heavy-atom pair within 8.0 Å and sequence separation $|i - j| \geq 4$.

Table 3: Contact density by chain length across 117 chains. [measured]

Chain length	n	median L	contacts/nt	map density
32–100	38	56	1.784	6.523%
100–200	6	124	2.851	4.043%
200–500	3	393	4.216	2.151%
500–1500	9	1038	4.532	0.974%
≥ 1500	61	2901	4.854	0.353%
Overall	median 4.397, p95 5.422, max 5.497			median 0.389%

Contacts per nucleotide saturates at 4.4–4.9 and is bounded by RNA coordination geometry, independent of length; density falls approximately as $1/L$. The number of contacts therefore grows *linearly* in L .

The consequence for architecture is immediate. A dense pair representation of dimension $c = 128$ requires, for 48 blocks holding roughly six intermediate tensors each and without gradient checkpointing:

$$L = 512 \rightarrow 19.3 \text{ GB}, \quad L = 1024 \rightarrow 77.3 \text{ GB}, \quad L = 2048 \rightarrow 309.2 \text{ GB}.$$

A dense pair track is not merely wasteful at RNA lengths; it is infeasible. Retaining $K = cL$ candidate edges with $c = 32$ costs 1.56% of dense at $L = 2048$ (65,536 edges, 16.8 MB per tensor) while providing roughly sixfold headroom over the measured p95 requirement of 5.42 contacts per nucleotide.

3.2 Ion coordination and local rigidity are one phenomenon

We z-normalise crystallographic B-factors within each structure so that refinement protocols are comparable, and restrict this analysis to the 31 X-ray structures among the sample; cryo-EM atomic displacement parameters are not comparable and were excluded. Lower z_B indicates greater rigidity.

Table 4: Normalised B-factor versus distance to the nearest Mg^{2+} , over 24,623 nucleotides. **[measured]**

Distance to nearest Mg^{2+}	n nucleotides	mean z_B
0–4 Å	319	− 0.883
4–6 Å	2,441	−0.673
6–8 Å	7,590	−0.415
8–12 Å	8,115	+0.004
12–20 Å	2,984	+0.756
> 20 Å	3,174	+ 0.877

The gradient is monotonic and spans 1.76 standard deviations. Local packing density independently correlates with rigidity at Pearson $r = -0.294$.

Architectural consequence. Ion effects and motif rigidity, which the project brief treats as two separate physical ingredients, are strongly coupled. A model that predicts Mg^{2+} sites obtains a rigidity map nearly for free and vice versa. That mutual information justifies routing them into a *shared* physics expert rather than two disconnected heads.

Caveats. B-factor also absorbs resolution, occupancy and refinement choices. Mg^{2+} proximity and dense packing are themselves correlated, since both mark folded cores, so this is association rather than isolated causation; the gradient partly restates that structured cores both bind ions and are ordered. The closest bin contains only 319 nucleotides. This is strong evidence for *using* the coupling as a prior, not a proof of mechanism.

3.3 Ion inventory and coordination chemistry

Table 5: Cation inventory across 179 parsed structures. **[measured]**

Ion	Count	Inner-sphere partner	Count
Mg^{2+}	17,428	OP2 (phosphate)	5,890
K^+	1,868	OP1 (phosphate)	5,144
Zn^{2+}	359	O6 (guanine)	757
Na^+	161	O4 (uracil)	511
Ca^{2+}	29	N7 (purine)	447
Mn^{2+}	12	O2' (ribose)	196

Mg^{2+} outnumbered all other cations by roughly 9:1, so an ion-aware model that handles a single species should handle Mg^{2+} . Some 83% of direct coordination is to the two non-bridging phosphate oxygens; the next largest class, G(O6)/G(N7), is the textbook major-groove guanine site.

Coordination is predominantly water-mediated: a mean of 0.77 inner-sphere against 7.00 outer-sphere RNA contacts per Mg^{2+} . Crucially the inner-sphere fraction is itself a function of

tertiary compactness: 0.511 overall but only 0.296 for chains shorter than 500 nt. The inner/outer ratio is therefore a *predictable readout* of compactness, not merely an input.

Stratification warning. These aggregates are ribosome-dominated and must not be quoted flat. The five largest structures hold 39.4% of all Mg^{2+} . Per structure the median is 72.5 ions against a mean of 181.5; per nucleotide the median is 0.046, roughly one Mg^{2+} per 22 nucleotides.

What crystallography cannot see. Resolved Mg^{2+} are site-bound ions only. The diffuse Manning atmosphere, which neutralises roughly 76% of backbone charge, is invisible to crystallography. The two channels must be modelled separately and never conflated.

3.4 A negative result that constrains the design

We tested whether GNRA-like tetraloop sequence context alone predicts rigidity, matching the 4-mers {GAAA, GCAA, GAGA, GUGA, GGAA, GCGA, GUAA, GAAG} anywhere in a chain.

Table 6: GNRA k -mer context versus rigidity. [measured] Effect size 0.073σ .

Context	n nucleotides	mean z_B
GNRA-like 4-mer	4,944	−0.082
All other	95,768	−0.009

The effect is negligible. The likely cause is methodological: a GNRA *sequence* is a GNRA *tetraloop* only when it actually caps a hairpin, and linear 4-mer matching counts mostly helical and junction occurrences.

Architectural consequence. Sequence context alone does not identify a motif. Motif identity requires the interaction graph. The motif machinery must therefore be conditioned on the pair representation after secondary structure is available, never on raw k -mers. This rules out the naive “learn motifs from sequence n -grams” design.

4 The PHAROS architecture

4.1 Inputs

Input	Required	Note
Sequence, L tokens, vocab 5	yes	elDORS is DNA-alphabet; $T \rightarrow U$ at load
Ionic condition c_{ion}	yes, defaulted	($[\text{Mg}^{2+}]$, $[\text{K}^+]$, $[\text{Na}^+]$, T , pH); novel
MSA / Rfam alignment	optional	graceful degradation when shallow
Chemical probing profile	optional	not yet acquired (Section 5)

Ionic condition enters as a FiLM-style conditioning vector, analogous to a diffusion timestep embedding. The same sequence at 0 mM and 10 mM Mg^{2+} must produce different predictions; this is the clearest novelty claim in the design.

4.2 Token trunk: hybrid attention with MoE

The trunk uses 32 blocks at $d = 768$ with the repeating pattern

$$[\text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{FULL}],$$

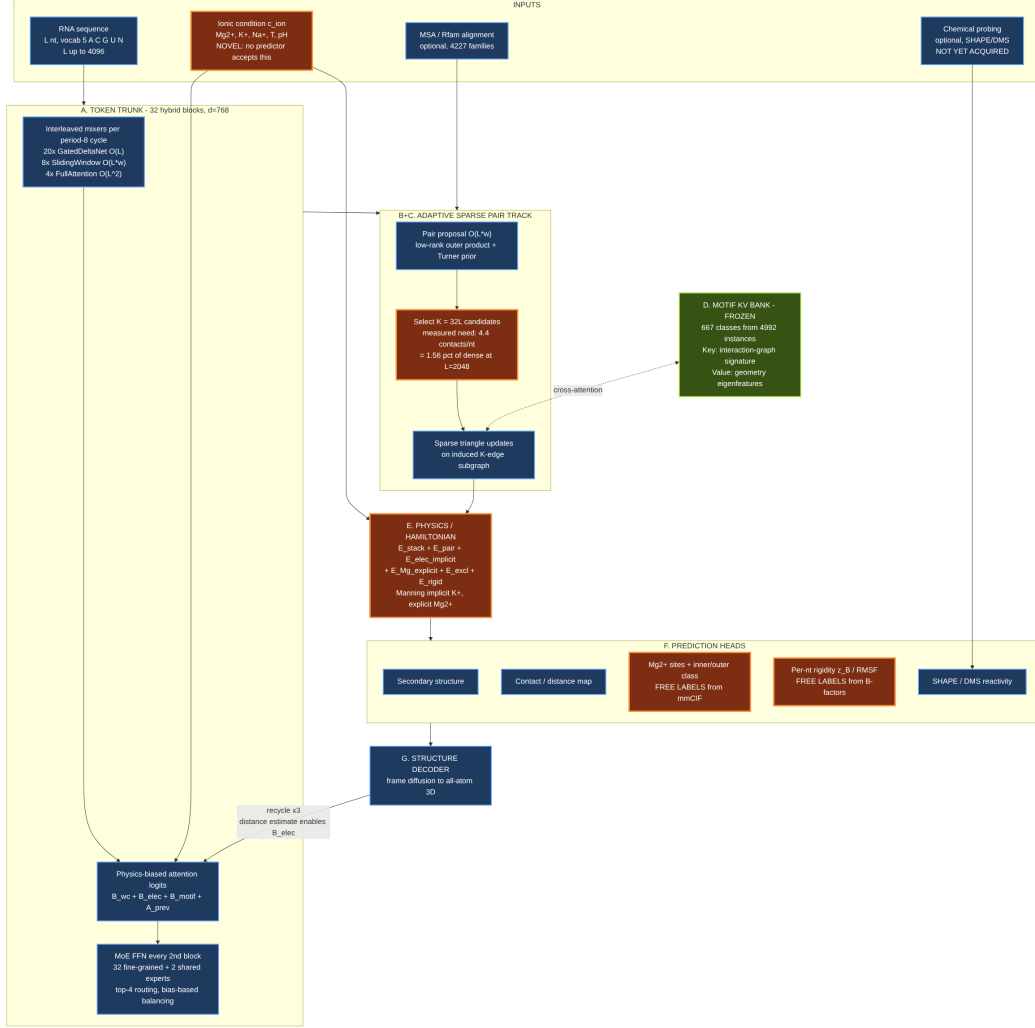


Figure 1: PHAROS system overview. Orange denotes novel components, green denotes frozen retrieval, blue denotes adapted standard components.

where GDN is Gated DeltaNet linear attention ($O(L)$, 20 of 32 blocks), SWA is sliding-window softmax attention with window 128 ($O(Lw)$, 8 blocks), and FULL is full attention with physics bias ($O(L^2)$, only 4 blocks).

The justification is both architectural and biological. Jamba interleaves Mamba, attention and MoE at a 7 : 1 ratio and reaches 256K context with a 4 GB KV cache; Gated DeltaNet outperforms Mamba2 and DeltaNet on in-context retrieval and length extrapolation; Oryx varies the mixer type *within* a sequence. RNA has exactly the structure this exploits: long stretches of regular A-form helix that are locally periodic and low-information, punctuated by rare junctions, pseudoknots and kissing loops that require global mixing. At $L = 4096$, an all-dense 32-block trunk would perform 537M pair evaluations; PHAROS performs 67M, an $8\times$ reduction.

4.2.1 Physics-biased attention

In SWA and FULL blocks the attention logits carry four additive terms:

$$A_{ij} = \frac{Q_i K_j^\top}{\sqrt{d}} + B_{wc}(i, j) + B_{elec}(i, j; c_{ion}) + B_{motif}(i, j) + A_{ij}^{(l-1)}. \quad (2)$$

B_{wc} generalises ERNIE-RNA’s fixed prior to a learned pair-type embedding modulated by stacking context, since real pairing energy is nearest-neighbour dependent. It is initialised at

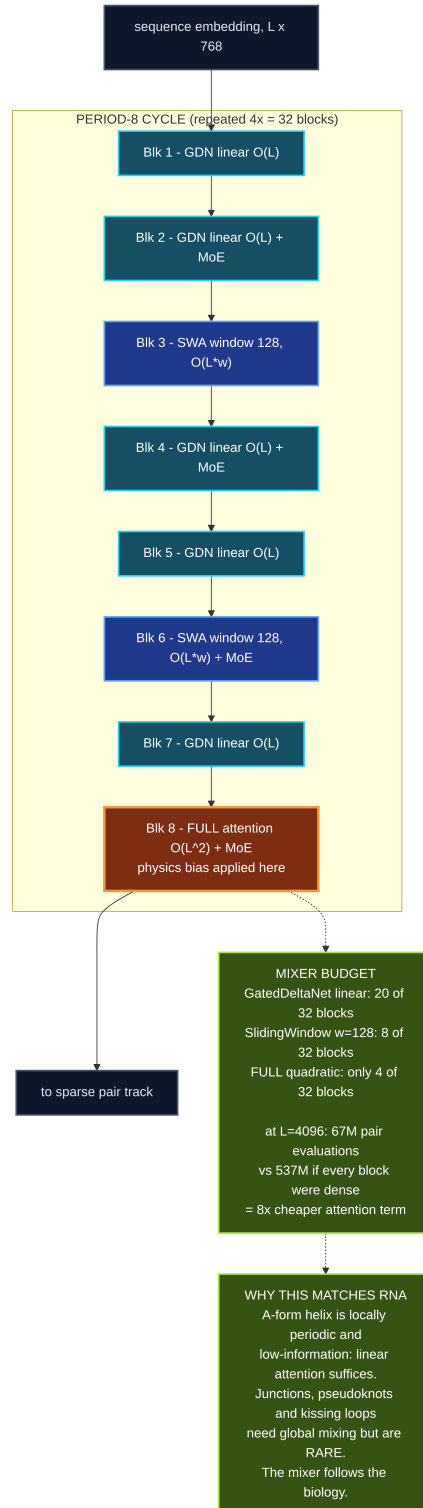


Figure 2: Period-8 mixer interleaving, repeated four times for 32 blocks.

the ERNIE-RNA values so training begins from a known-good prior. The term $A^{(l-1)}$ retains ERNIE-RNA’s attention accumulation.

B_{elec} is the novel term. With Bjerrum length $\ell_B = e^2/(4\pi\epsilon_0\epsilon_r k_B T) \approx 7.1 \text{ \AA}$ in water at 298 K, Manning parameter $\xi = \ell_B/b$ for axial charge spacing b , condensed fraction and renormalised charge

$$\theta = 1 - \frac{1}{z\xi}, \quad q_{\text{eff}} = -(1 - \theta), \quad (3)$$

and Debye screening parameter $\kappa = \sqrt{2N_A e^2 I / (\epsilon\epsilon_0 k_B T)}$ from ionic strength I , the bias is

$$B_{\text{elec}}(i, j) = -\lambda q_{\text{eff}}^2 \ell_B \frac{\exp(-\kappa d_{ij})}{d_{ij}}. \quad (4)$$

For monovalent RNA $\theta \approx 0.76$, so roughly three quarters of backbone charge is screened before any structure-specific electrostatics is considered. Only λ is learned; q_{eff} , κ and ℓ_B follow in closed form from c_{ion} and temperature. Equation (4) is the mechanism by which salt concentration changes the prediction.

Resolving the circularity. Equation (4) requires d_{ij} , which is unknown on the first pass. B_{elec} is therefore disabled at recycle 0 and enabled from recycle 1 using the current predicted distance map. Recycling is thus structural to the design rather than an optimisation: the physics term needs a geometry estimate to evaluate, and each recycle sharpens both.

4.2.2 Mixture-of-experts feed-forward

Following DeepSeekMoE, the design uses 32 fine-grained routed experts at $d_{\text{ff}} = 512$ plus 2 always-on shared experts, with top-4 routing giving 6 active experts of 34, in every second block.

Shared-expert isolation matters for a specific reason here: Watson–Crick helix formation is the highest-frequency pattern in all RNA. Placing it in an always-on shared expert prevents all 32 routed experts from separately relearning it, freeing them to specialise on what is rare and hard—junctions, pseudoknots and ion pockets.

Load balancing is auxiliary-loss-free, adjusting a per-expert bias each step rather than adding a loss term. This matters acutely here because our 3D data is pathologically imbalanced: 5S rRNA alone accounts for 5,976 BGSU clusters, so an auxiliary balance loss would actively fight the biology.

The router input is the token hidden state concatenated with the current local structural estimate—pairing probability, predicted rigidity, local density. This follows directly from the GNRA negative result of Table 6: routing on sequence k -mers does not work. It also prevents router collapse onto rRNA, because structural regimes are far more evenly distributed across the corpus than families are.

4.2.3 Parameter budget

PHAROS-Base carries $2.3\times$ the capacity of NucleicBERT at slightly lower active compute, with an attention term roughly $8\times$ cheaper at long context.

4.3 Adaptive sparse pair track

The track proceeds in four steps. A proposal stage computes candidate pair scores in $O(Lw)$ from a low-rank outer product plus a Turner nearest-neighbour prior, never materialising an $L \times L$ tensor. A selection stage retains $K = 32L$ candidates. Sparse triangle updates then enforce the triangle inequality on the induced subgraph. Finally candidates are re-scored after every recycle, making the top- K a soft, revisable commitment rather than a hard early prune.

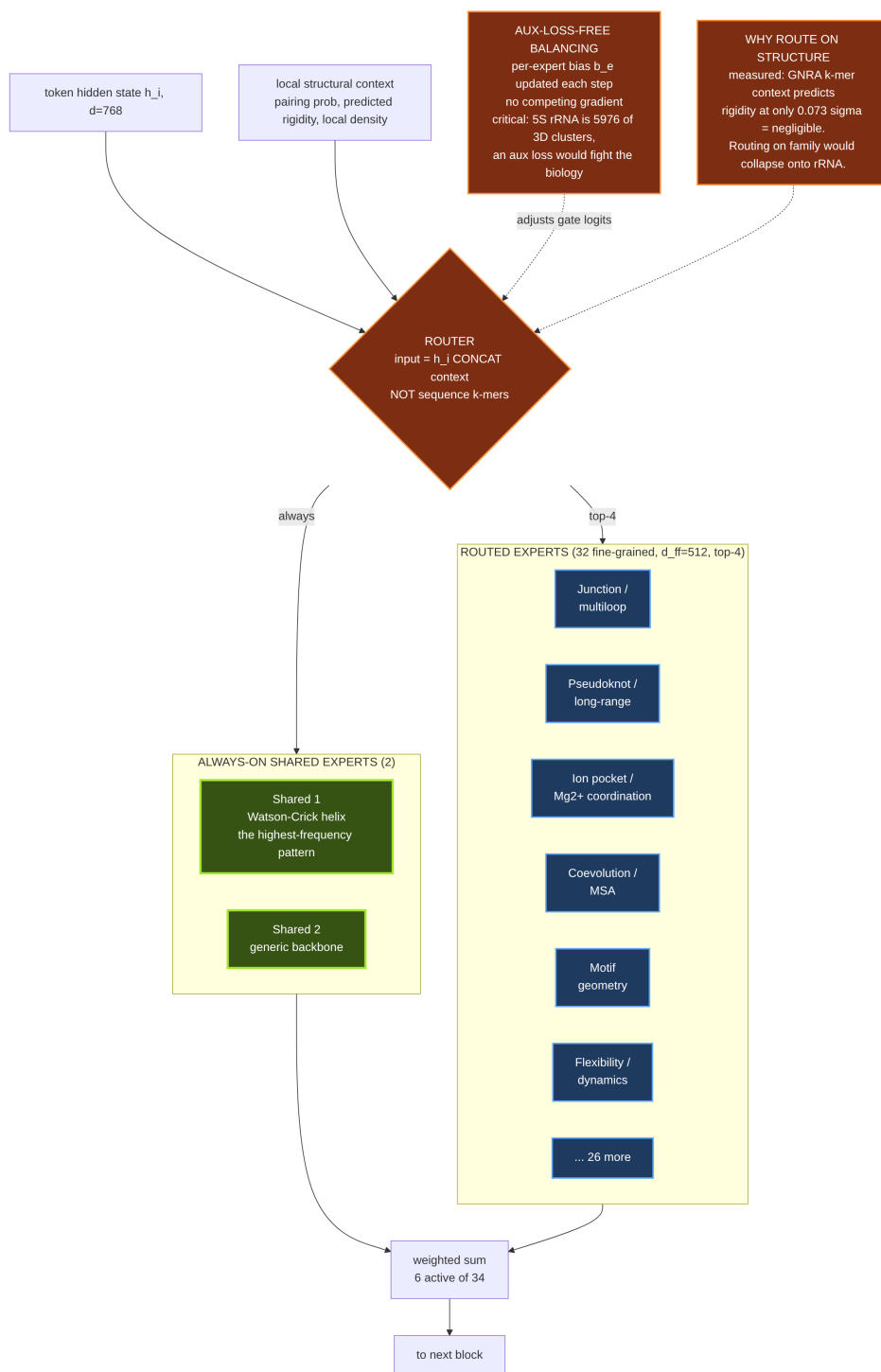


Figure 3: Expert routing. Routing keys on local structural regime, not sequence.

Table 7: PHAROS-Base parameter budget against NucleicBERT.

Component	Total	Active
Embeddings (vocab 5)	0.01M	0.01M
Attention, 32 blocks, $d = 768$	75.5M	75.5M
MoE FFN, 16 blocks $\times$ 34 experts	642M	113M
Dense FFN, 16 non-MoE blocks	113M	113M
Sparse pair track and triangle updates	45M	45M
Motif bank, heads, decoder	35M	35M
PHAROS-Base total	$\sim 911\text{M}$	$\sim 382\text{M}$
NucleicBERT (dense)	404M	404M

Failure mode. A contact never proposed can never be predicted. The mitigations are the sixfold headroom of Table 3, per-recycle re-selection, and a recall-oriented auxiliary loss on the proposal stage. **Proposal recall must be reported per length bin.** A sparse track that silently drops five percent of true contacts would impose an invisible accuracy ceiling, and must be measured rather than assumed.

4.4 Motif retrieval bank

The RNA 3D Motif Atlas 4.12 compresses 4,992 loop instances into 667 motif classes, 413 internal and 254 hairpin: a 7.5:1 ratio indicating that RNA tertiary structure is far more repetitive than its sequence diversity suggests. PHAROS precomputes a frozen key–value bank over these classes, with keys encoding the non-canonical interaction-graph signature and values encoding internal geometry as distance-matrix eigenfeatures plus a backbone torsion profile. The pair track cross-attends into this bank.

This is the appropriate response to data scarcity. With only 6,661 unique sequences carrying 3D structure, a model cannot reliably learn motif geometry implicitly. The bank supplies the geometry explicitly, leaving the model to learn only *where* motifs occur—a substantially easier problem than learning what they look like. The bank is roughly 10^3 fixed entries shared across a batch, so its cost is negligible. Consistent with Table 6, retrieval is keyed on the interaction graph, never on sequence k -mers.

4.5 Physics module

A differentiable energy is evaluated on the predicted geometry:

$$E_{\text{total}} = E_{\text{stack}} + E_{\text{pair}} + E_{\text{elec}}^{\text{implicit}} + E_{\text{Mg}}^{\text{explicit}} + E_{\text{excl}} + E_{\text{rigid}}. \quad (5)$$

The implicit/explicit split follows STEM, which establishes that mean-field theory captures monovalent KCl well but fails for Mg^{2+} , where ion–ion correlation and site-specific chelation dominate. Our own inventory agrees on which species matters: Mg^{2+} outnumbers K^+ by 9:1.

Inner/outer-sphere classification is a first-class output because STEM’s central claim is that folding dynamics are driven by *exchange between* direct and water-mediated coordination—the mode, not the presence, is decisive. Our measurements supply both channels directly and show the inner fraction varies with compactness (0.511 overall, 0.296 below 500 nt).

The term E_{rigid} closes the loop with Table 4: predicted Mg^{2+} sites raise local rigidity, which stiffens harmonic restraints, which constrains the decoder. That pathway is the architectural encoding of the measured 1.76σ gradient.

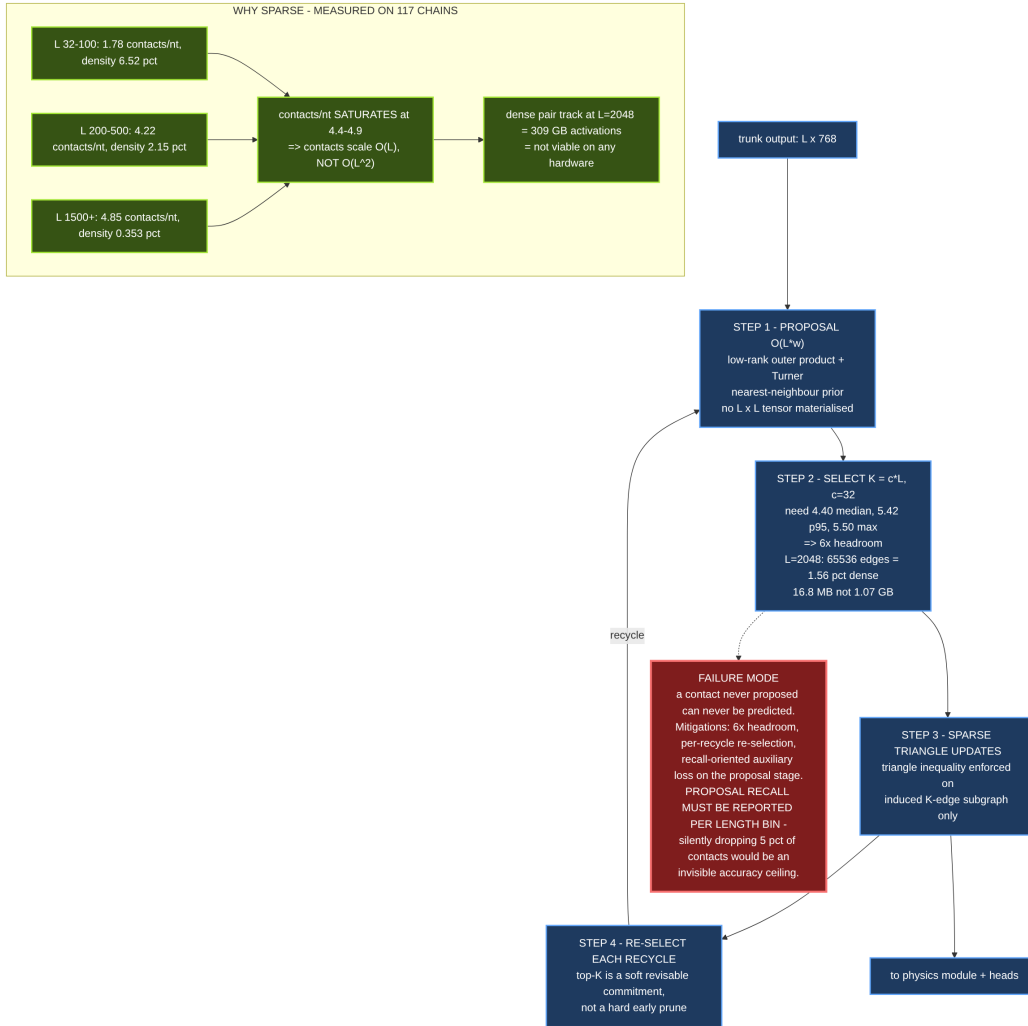


Figure 4: The sparse pair track and its measured justification.

4.6 Heads and decoder

The decoder is frame-based diffusion. We retain an explicit geometric and energetic stage rather than going fully generative, consistent with the CASP16 observation that physics-plus-ML hybrids currently lead in RNA and with trRosettaRNA’s restraint-then-fold approach outscoring AlphaFold3’s raw-atom diffusion.

Mg^{2+} sites and B-factor rigidity deserve emphasis: both are already present inside every mmCIF file the field routinely downloads, are used by no current structure predictor, and were extracted end-to-end in preparing this report.

5 Data mapping and the acquisition gap

Six of seven training stages are fully served by the existing catalogue. The exception is chemical probing, and it is the most valuable missing asset. Ribonanza provides 2.1M sequences with DMS and SHAPE measurements—roughly $315\times$ more supervised examples than the 6,661 unique sequences with 3D structure. It is the only label source that is simultaneously abundant, structure-informative and already public. This is the highest-priority acquisition.

Corpus weighting. The elDORS chunks are source-partitioned rather than homogeneous: 9 of 20 chunks have a median length of exactly 151 nt, indicating unassembled sequencing reads, while

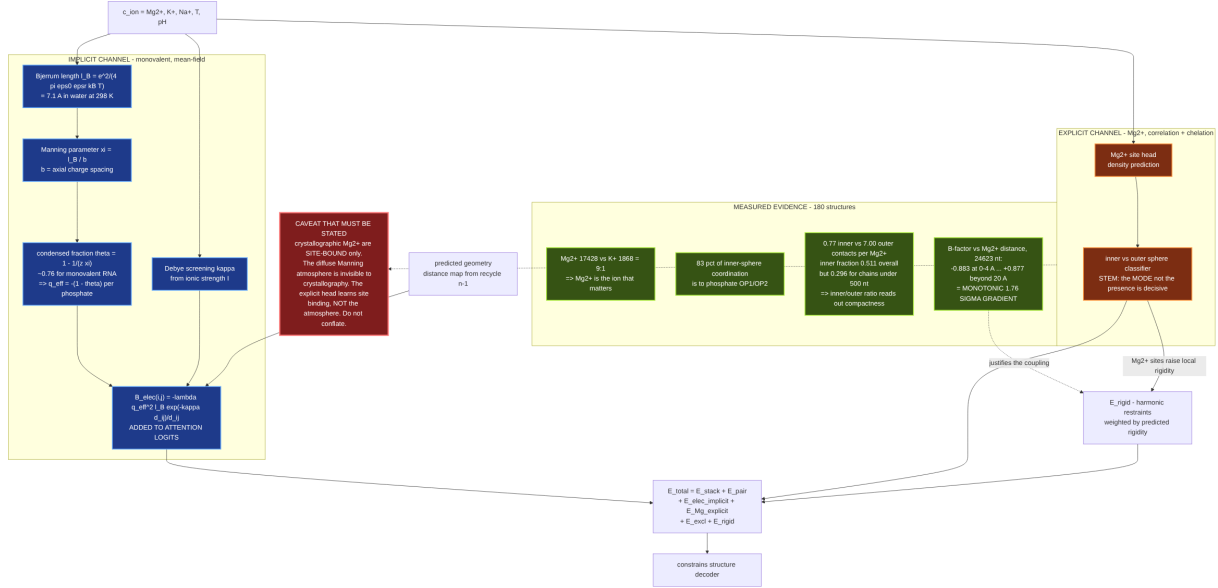


Figure 5: The hybrid implicit/explicit electrostatic treatment.

Table 8: Prediction heads and their supervision. Two channels are effectively free.

Head	Supervision source	Status	Labels
Secondary structure	bpRNA, ArchiveII, RNAstrAlign	in catalogue	~160k
Contact / distance	RNA3DB, gRNade	in catalogue	27,452 chains
Mg^{2+} sites	extracted from mmCIF	free	17,428 (180 structs)
Rigidity (z_B , RMSF)	B-factors in mmCIF	free	every X-ray struct
Reactivity (SHAPE/DMS)	Ribonanza / RMDB	not acquired	—
3D coordinates	RNA3DB	in catalogue	6,661 unique seqs

chunks 6 and 9 (median 2,193–2,389 nt) are the only substantial source of long-context signal. Uniform sampling in Stage 1 would allow read fragments to dominate pretraining, so per-chunk weighting is required.

6 Novelty claims

7 Risks and open questions

1. **Proposal recall ceiling.** If the sparse proposal stage misses true contacts, nothing downstream recovers them. Must be measured per length bin.
2. **Ionic metadata sparsity.** mmCIF records crystallisation conditions inconsistently. Stage 5 may have far fewer usable ionic labels than hoped. This requires an audit before committing to that stage, and is currently **open**.
3. **Crystallographic ions are not the solution ensemble.** The explicit head learns site binding, not the diffuse atmosphere. These must not be conflated.
4. **B-factor is a noisy rigidity proxy.** Excluding cryo-EM left only 31 of 137 parsed structures usable, which materially shrinks the rigidity label set.
5. **Association, not isolated causation.** Mg^{2+} proximity and packing density both track folded cores.

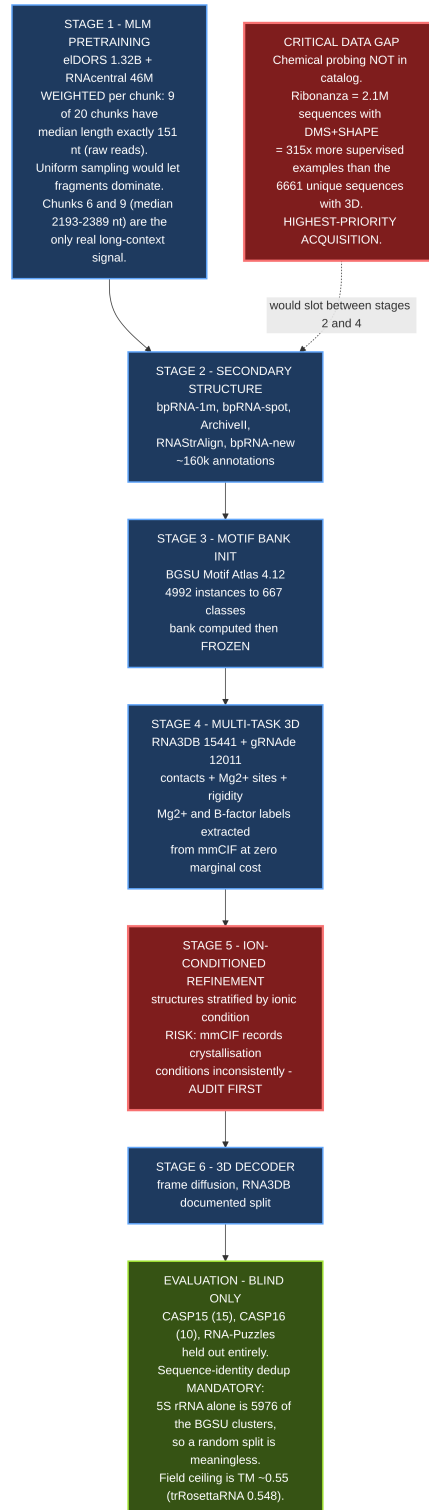


Figure 6: Training curriculum, data sources, and the chemical-probing gap.

Table 9: Novelty claims stated precisely, with prior-art status.

Claim	Prior-art status
Ionic conditions as a model input, reaching attention logits through a Manning-screened Coulomb bias	No existing RNA structure predictor accepts ionic conditions.
First sparse-MoE architecture for RNA structure	AIDO.Protein is MoE for protein; no RNA equivalent exists.
$O(L)$ sparse pair track justified by measured contact scaling	AlphaFold3 and Rhoformer are dense $O(L^2)$.
Frozen motif KV bank as a retrieval-based geometry prior	The Motif Atlas is published but wired into no large model.
Mg ²⁺ sites and B-factor rigidity as free auxiliary supervision	Present in mmCIF; used by no structure predictor.
Coupled ion-rigidity expert justified by a measured 1.76σ gradient	Treated separately or not at all elsewhere.

6. **Infrastructure.** NucleicBERT used 192 A100s densely. MoE improves FLOP efficiency but adds routing and communication complexity.
7. **Evaluation honesty.** The field ceiling is TM ≈ 0.55 . Any improvement claim must be made on blind sets—CASP15/16 and RNA-Puzzles—never on rRNA-saturated random splits. Sequence-identity deduplication is mandatory.

8 Status

This is a design specification. No model has been trained and no predictive performance is claimed. What has been established is the evidentiary basis: the contact-scaling law, the ion-rigidity coupling, the ion inventory and coordination chemistry, and the GNRA negative result, all measured on 180 non-redundant structures with reproducible scripts. The next phase is to implement the sparse pair track and validate proposal recall before any large-scale training is attempted.

A Reproducing the measurements

Script	Output
<code>scripts/sampling/fetch_samples.py</code>	180 mmCIF, 12 Rfam seeds
<code>scripts/sampling/analyze_ions_motifs.py</code>	<code>ion_summary.json</code>
<code>scripts/sampling/analyze_rigidity.py</code>	<code>rigidity_summary.json</code>
<code>scripts/sampling/analyze_contact_sparsity.py</code>	<code>contact_sparsity.json</code>

References

- [1] ERNIE-RNA: an RNA language model with structure-enhanced representations. *Nature Communications*, 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12627772/>
- [2] Accurate RNA 3D structure prediction using a language model-based deep learning approach (RhoFold+). *Nature Methods*, 2024. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11621015/>

- [3] Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11168924/>
- [4] Structural Topology-based Electrostatic Model (STEM) Reveals Ion-Coordination Exchange as a Driver of RNA Folding Dynamics, 2026. <https://pmc.ncbi.nlm.nih.gov/articles/PMC13345221/>
- [5] Generalized Manning Condensation Model Captures the RNA Ion Atmosphere, 2015. <https://pubmed.ncbi.nlm.nih.gov/26197147/>
- [6] The RNA 3D Motif Atlas: Computational methods for extraction, organization and evaluation of RNA motifs. *Methods*, 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4921307/>
- [7] Ribonanza: deep learning of RNA structure through dual crowdsourcing, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10925082/>
- [8] DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models, 2024. <https://arxiv.org/html/2401.06066v1>
- [9] DeepSeek-V3 Technical Report, 2024. <https://arxiv.org/pdf/2412.19437>
- [10] Gated Delta Networks: Improving Mamba2 with Delta Rule. *ICLR*, 2025. <https://arxiv.org/pdf/2412.06464>
- [11] Mixture of Experts Enable Efficient and Effective Protein Understanding and Design (AIDO.Protein). *bioRxiv*, 2024. <https://www.biorxiv.org/content/10.1101/2024.11.29.625425v1>
- [12] Deep learning for RNA structure prediction. *Current Opinion in Structural Biology*, 2025. <https://www.sciencedirect.com/science/article/pii/S0959440X25000090>
- [13] Enhancing RNA 3D Structure Prediction in CASP16: Integrating Physics-Based Modeling with Machine Learning, 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12354339/>
- [14] Elastic network models for RNA: a comparative assessment with molecular dynamics and SHAPE experiments. *NAR*, 2015. <https://academic.oup.com/nar/article/43/15/7260/2414435>
- [15] Graph deep learning locates magnesium ions in RNA, 2023. <https://pubmed.ncbi.nlm.nih.gov/37292390/>
- [16] Comprehensive benchmarking of large language models for RNA secondary structure prediction. *Briefings in Bioinformatics*, 2025. <https://academic.oup.com/bib/article/26/2/bbaf137/8109668>